Classroom materials

Use these ready-to-print pages with the student workbook, teacher guide and the physical parts. No paid lesson materials or course subscriptions are needed.

Before the first class

Open the course, charge and connect the kit, and prepare one checked robot base per team. The teacher still needs to test the physical builds and the programs on the actual kit.

What to print

For each team of four	Where to find it
One set of role cards	This pack
One set of arrow and color cards	This pack
Paper parcel and attachment templates	This pack; choose only the final build needed
One build worksheet for the current lesson	Each lesson’s Build slide
Student test notes	Student workbook
The current lesson’s coding guide	This pack or the online Coding guide link

Physical supplies per team

One LEGO #45522 kit, a compatible computer or tablet, a ruler, pencil, scissors, removable tape, white/red/blue paper and light cardboard. The drawing choice needs a washable marker, large paper and floor protection. The sweeper needs three tissue squares. The ball choice uses one crumpled paper ball. The rescue choice uses a light paper figure.

Teacher supplies

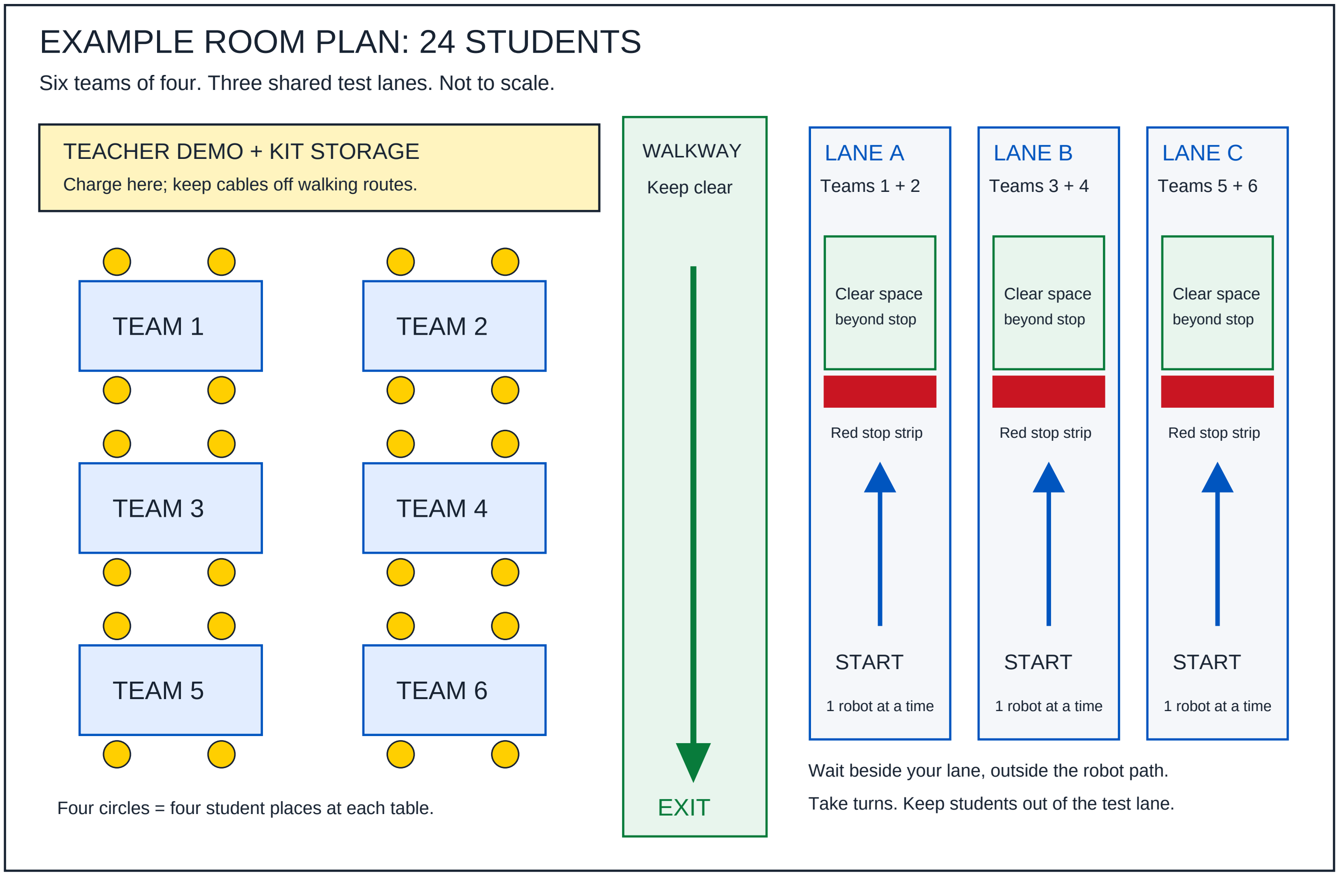
Charging access, a display for demonstrating the slides, a printer or screen access to these materials, firm supports for raised-wheel tests, and a clear floor area. Use the official instruction book supplied for the kit’s base build.

Choose your first session

Start with module 1. Use the room plan, hand out role cards, and show one color-to-message test before teams try. All student reading, worksheets and example answers are provided in the course.

Set up the classroom

An example for six teams of four. Move the tables and lanes to fit your room. Keep a clear route to the real exit.



Before testing

Make every lane at least twice as wide as the robot. Leave clear floor beyond each stop strip. Decide the actual lane length using the teacher's slow-drive test. The picture gives zones, not room measurements.

Who uses each lane?

Teams 1 and 2 share lane A. Teams 3 and 4 share lane B. Teams 5 and 6 share lane C. Only one robot uses each lane at a time. The other team builds, writes notes or waits outside the lane.

Where do students stand?

The test leader stands beside the lane at the Stop control. Observers stay outside the taped boundary. Turn power off before carrying the robot back. Keep charging cables on the storage table.

In the online Teacher Desk, hover over, focus on or tap "See an example classroom setup" to open this plan.

Your first 45-minute class

Use this script after the teacher has built and checked the base and made the lesson 1 program.

Time	Say or do	Students do
0–5 min	“Our robot will read a color and show a message. Today the wheels stay still.” Show the goal and word help.	Name a color; predict a message.
5–10 min	Hand out roles. Point to Stop and power off. Show the raised-wheel supports.	Each student explains their role and the Stop method.
10–15 min	Show white, red and blue paper. Read the rule: white READY; red DELIVERY; other CHECK.	Say the expected result before each card.
15–30 min	Open module 1’s coding guide. Show one step at a time; use the prepared program when a team needs a starting point.	Connect their own kit with help. Try all three cards and record every result.
30–38 min	“Change one screen message. What should happen now?”	Change one message and repeat the color test.
38–42 min	Ask: “What went in? What came out?”	Explain color input and screen-message output.
42–45 min	“Save your code, power off, count the pieces and return the cards.”	Save, tidy and leave their notes with the team kit.

If a group gets stuck

Show one card at a time. Keep the sensor and lighting still. Ask students to point to the rule before running it. Offer the prepared code and ask them to change one message. Do not start driving in this lesson.

Use the Mission simulator

The moving brick robot is a screen activity. Use it to teach planning, instructions in order and learning from a failed try. The real kit uses floor colors and still needs its own tests.

Before students begin

Choose a challenge. Ask students to write a prediction before running. Add instructions, then choose One step to watch each move or Run to play the route. Blue squares are barriers. The yellow star marks the delivery space.

Three stars, three clear goals

Star	What the student does
Delivery	Ends the program with the robot on the delivery space.
Route planning	Delivers in eight steps or fewer in challenges 1–2, or six or fewer in challenge 3. A longer route can still earn delivery credit.
Test recording	Writes a prediction before the run, then saves a successful test without rewriting that prediction afterward.

When the robot bumps

The robot stops and a message explains the blocked route. Ask: “Which instruction caused the problem?” “What one change will you try?” Students keep earlier stars. There are no lost lives, countdowns or penalties for retrying.

Sound and movement

Sound starts off. Turn it on with the Sound button when it helps, or leave it off for a quiet room. Every sound has a visible message. The robot moves and turns between squares; the page respects the device’s reduced-motion setting.

Collect evidence of learning

Stars show game progress, not a physical-robot grade. Ask students to explain one decision and one revision. Use Save test result and Download test notes before refreshing or closing the page. Stars and notes do not carry over to a new page session.

The three checked example routes

F means Forward 1. R means Turn right. I means If blocked.
First delivery: F F F R F F F F.
Around the corner: F R F F F F R F.
Sensor decision: F I F F F F.

Team role cards

Cut out the four cards. Each student takes one job. Swap after a test so everyone builds, codes, starts and stops, and records results.

BUILD

Keep power off when changing parts.

Follow one build step at a time.

Check that wheels and controls are clear.

Swap jobs after each test.

CODE

Read each program step aloud.

Make the code and save it with a name.

Explain what each part does.

Swap jobs after each test.

START + STOP

Ask the teacher to check the setup.

Clear the lane and run one test.

Stay ready at Stop; turn off before reset.

Swap jobs after each test.

TEST NOTES

Write the prediction before running.

Record every run, even one that fails.

Name one change using the results.

Swap jobs after each test.

Color-test cards

Cut around each square. Keep labels and shiny tape away from the area the sensor reads. Test your printer's colors on the real sensor; use plain colored paper if needed.

RED — 8 cm square

WHITE — 8 cm square

BLUE — 8 cm square

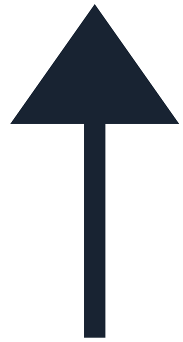
Print at Actual size / 100%. Check the 5 cm line below with a ruler. Do not use "Fit to page." Resize attachments if your actual base has less clear space.

5 cm

Route-planning cards

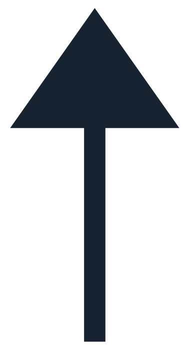
Cut out the cards. Arrange them before making the program. The arrows are a plan; the robot does not read them.

FORWARD

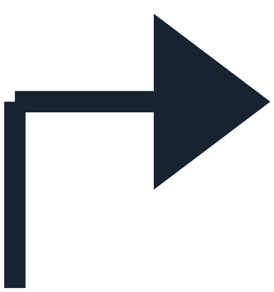


FORWARD

FORWARD



FORWARD

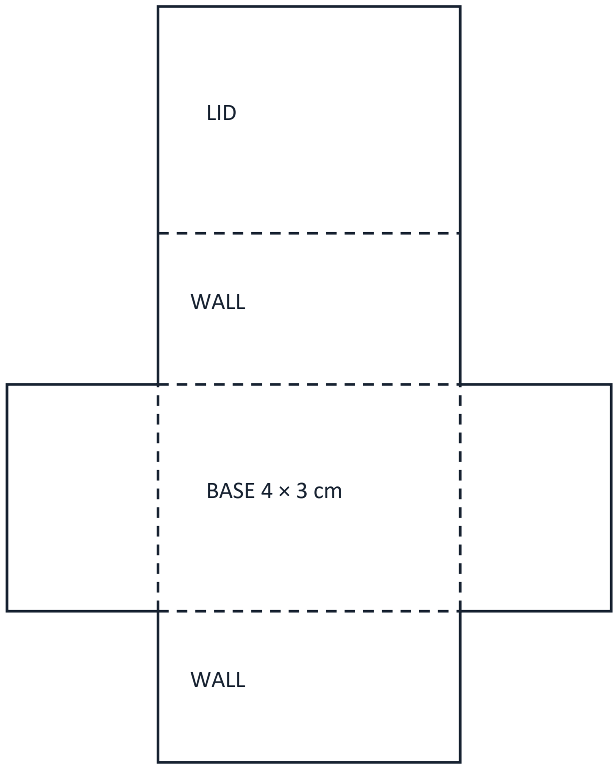


TURN RIGHT

STOP

Paper parcel

Cut the outside outline. Fold the dashed lines around the BASE. Fold up the four walls, bring the lid over and tape the edges. Keep it empty and light.



Print at Actual size / 100%. Check the 5 cm line below with a ruler. Do not use “Fit to page.” Resize attachments if your actual base has less clear space.

5 cm

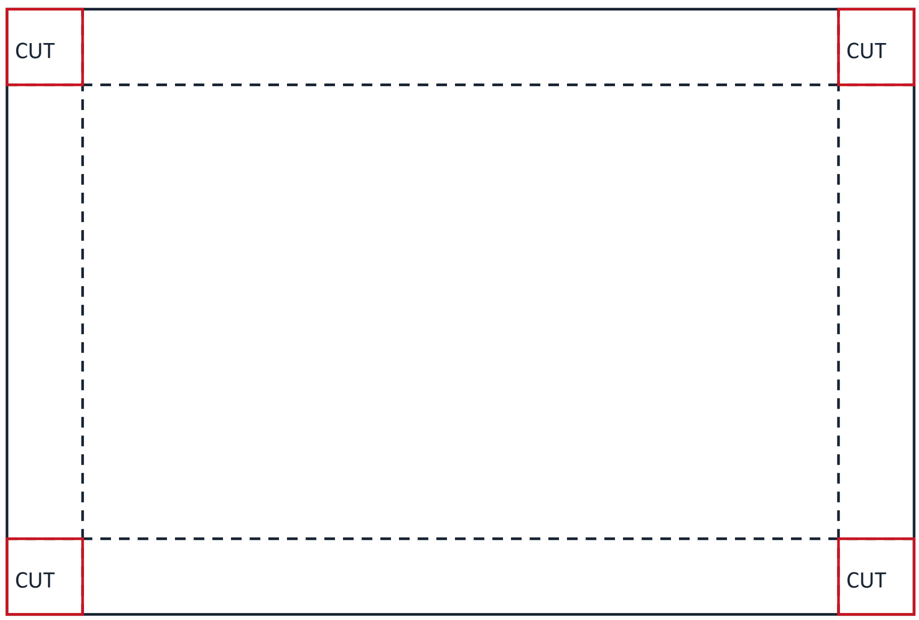
Teacher check

Measure the actual clear mounting space. Keep wheels, sensor and Stop controls clear. Turn the power off before adding or adjusting parts.

Parcel tray template

Cut out the four corner squares marked CUT. Fold the dashed lines up to make 1 cm walls. Tape the corners. Use the worksheet to attach the tray.

12 cm × 8 cm



Print at Actual size / 100%. Check the 5 cm line below with a ruler. Do not use “Fit to page.” Resize attachments if your actual base has less clear space.

5 cm

Teacher check

Measure the actual clear mounting space. Keep wheels, sensor and Stop controls clear. Turn the power off before adding or adjusting parts.

Marker sleeve template

Cut the rectangle and copy it onto thin card. Wrap it loosely around a washable marker and tape card to card. Attach and adjust it using the drawing-robot worksheet.

12 cm × 3 cm

Print at Actual size / 100%. Check the 5 cm line below with a ruler. Do not use “Fit to page.” Resize attachments if your actual base has less clear space.

5 cm

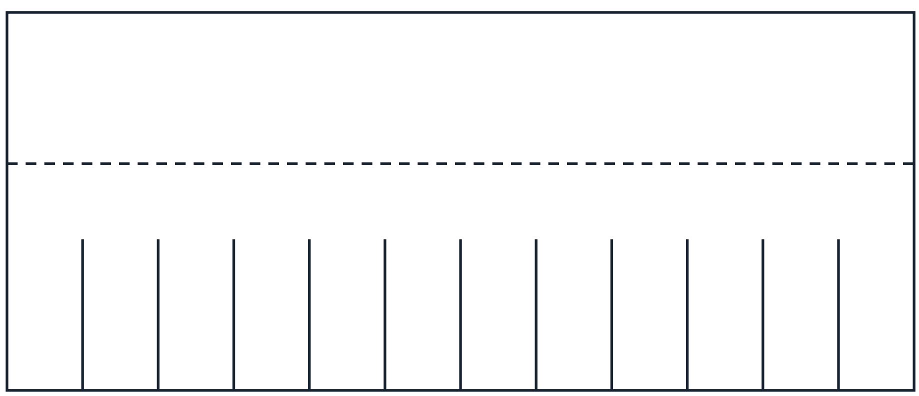
Teacher check

Measure the actual clear mounting space. Keep wheels, sensor and Stop controls clear. Turn the power off before adding or adjusting parts.

Paper sweeper template

Cut the rectangle. Fold the upper 2 cm strip back. Cut the short solid slits from the lower edge to make bristles. Keep the cuts below the fold. Follow the sweeper worksheet.

12 cm × 5 cm



Print at Actual size / 100%. Check the 5 cm line below with a ruler. Do not use “Fit to page.” Resize attachments if your actual base has less clear space.

5 cm

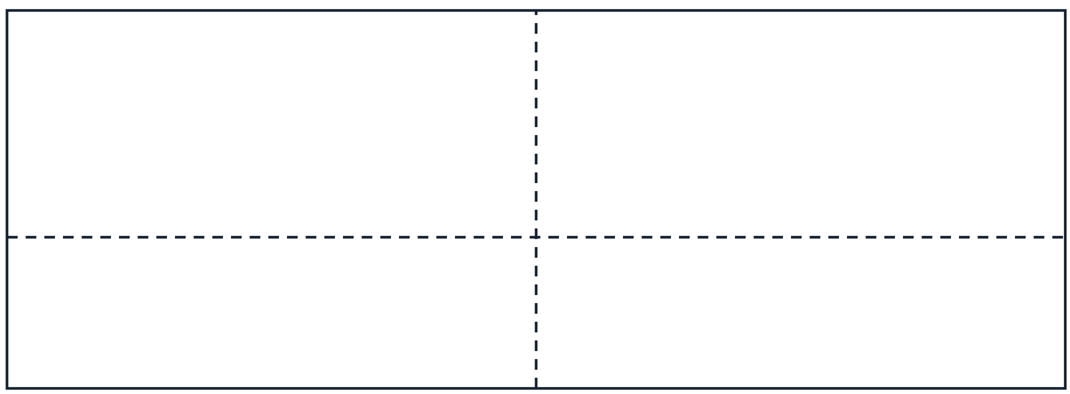
Teacher check

Measure the actual clear mounting space. Keep wheels, sensor and Stop controls clear. Turn the power off before adding or adjusting parts.

Ball guide template

Cut the rectangle. Fold the long dashed line to make a 2 cm mounting flap and a 3 cm upright wall. Fold gently at the middle to form a shallow V. Follow the ball-pusher worksheet.

14 cm × 5 cm



Print at Actual size / 100%. Check the 5 cm line below with a ruler. Do not use “Fit to page.” Resize attachments if your actual base has less clear space.

5 cm

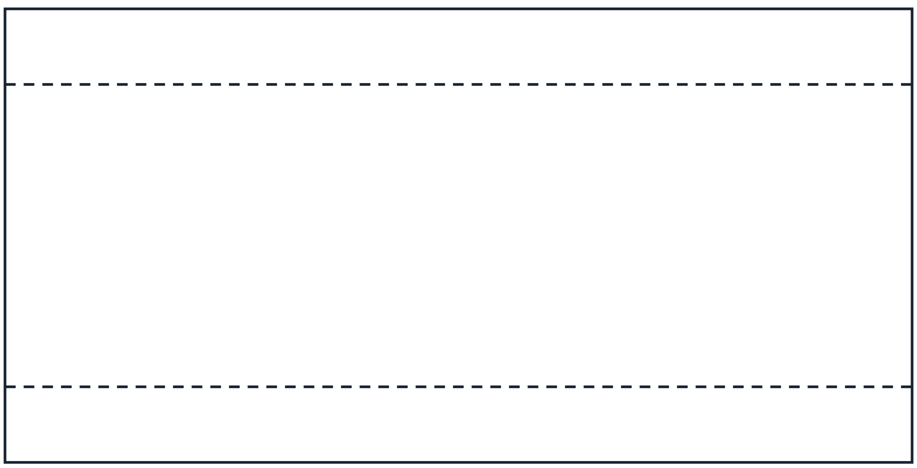
Teacher check

Measure the actual clear mounting space. Keep wheels, sensor and Stop controls clear. Turn the power off before adding or adjusting parts.

Rescue sled and tow bar

Cut the two rectangles. Fold the sled sides up at the dashed lines. Curl its front edge slightly. Cut the bands and fit them around a light paper figure. Follow the rescue-carrier worksheet.

12 cm × 6 cm



Cut two paper bands. Resize to fit your paper figure.

Print at Actual size / 100%. Check the 5 cm line below with a ruler. Do not use “Fit to page.” Resize attachments if your actual base has less clear space.

5 cm

Teacher check

Measure the actual clear mounting space. Keep wheels, sensor and Stop controls clear. Turn the power off before adding or adjusting parts.

Coding guides

The next pages give every lesson's written program steps. A program is a list of instructions. Use these directions in LEGO's free Coding Canvas. They are not automatic project files.

Before students use the kit

Follow the kit's connection instructions. Select only your team's parts. Check the direction of movement with raised wheels. Make and save each program, then test it on the actual robot before teaching.

Start small

Build one step or short section. Predict what it should do. Run it, check the result and stop before adding more. A physical robot needs a clear floor lane and someone ready at Stop.

When the app uses a different word

Match the action described here to the corresponding app block. Keep the same time limits, color rules and Stop actions. The current kit still needs a physical tryout; screen-game success does not check its hardware.

What students should save

Use a name such as Team2_Lesson6_Try1. Keep the previous working version before changing a setting. Record the change in the worksheet.

Lesson 1: Wake up your rover

Words to know

- Sensor: A part that detects something, such as a color.
- Program: The steps the robot follows.
- Input / output: The color reading goes in. A screen message comes out.

Coding and testing steps

1. Start with all movement stopped. Do not add any steps that move a motor. A motor is the part that makes a wheel turn.

2. Repeat the following color check 60 times. Repeat means do the same steps again.

3. Read the color. If it is white, show READY on the screen. If it is red, show DELIVERY. For any other reading, show CHECK.

4. Wait 0.5 seconds before the next color check.

5. Stop all movement again and end the program.

The checks last about 30 seconds. Hold one card steady until its reading appears. Run the program again if you need more time. Keep the wheels still.

Teacher check before running

Connect the correct team's robot. Check the screen Stop control and show how to turn the power off. Raise the wheels for the first check. Keep someone ready to stop the robot.
Keep the wheels still for this lesson. Test the colors before any driving lesson.

LEGO help

Build instructions: https://www.lego.com/service/buildinginstructions/45522

Coding app: https://code.legoeducation.com/

Teacher help: https://teach.legoeducation.com/

Lesson 2: Make a short drive

Words to know

- Motor: A part that makes something move.
- Bounded: Set to stop after a certain time or number of steps.
- Trial: One test run.

Coding and testing steps

1. Start with all movement stopped.
2. Set a low driving speed, such as 20 percent. Test forward direction with the wheels lifted first.
3. Move forward for 0.8 seconds. Use a timed movement block that stops, or add Stop immediately after the timed move.
4. Stop all movement and end the program.
5. On the floor, repeat this short run three times. Measure each distance. With your teacher, also test the Stop control during a drive.

Teacher check before running

Connect the correct team's robot. Check the screen Stop control and show how to turn the power off. Raise the wheels for the first check. Keep someone ready to stop the robot. Keep the lane clear and leave extra space beyond the finish.
Start slowly. Use a short time limit and check that the program stops by itself. Stop at once if it moves the wrong way.

LEGO help

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Lesson 3: Build a parcel holder

Words to know

- Cargo: The thing your robot carries.
- Cradle: A holder that keeps the parcel in place.
- Fair test: Change one thing and keep the rest the same.

Coding and testing steps

1. Use the same short driving program as lesson 2: low speed, 0.8 seconds forward, then Stop.

2. Turn off the robot before adding the parcel holder. Keep the wheels, sensor and Stop controls clear.

3. Use the same paper parcel for three test runs. Write down whether it stays on.

4. Turn off the robot. Change one part of the holder.

5. Repeat three runs with the same parcel, start line, floor and code. Compare the results.

Teacher check before running

Connect the correct team’s robot. Check the screen Stop control and show how to turn the power off. Raise the wheels for the first check. Keep someone ready to stop the robot. Keep the lane clear and leave extra space beyond the finish.
Start slowly. Use a short time limit and check that the program stops by itself. Stop at once if it moves the wrong way.

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Lesson 4: Help your robot read colors

Words to know

- Reading: The information a sensor gives you.
- Unknown reading: The sensor cannot name the color.
- Fair test: Change one thing at a time.

Coding and testing steps

1. Start with all movement stopped. Do not add any steps that move the wheels.
2. Repeat 60 times: read the color, show its name on the screen, and wait 0.5 seconds.
3. Stop all movement and end the program.
4. Use this program to test white, red and another paper color three times each. Write down all nine readings.
5. If a reading is wrong or unknown, keep the wheels still. Change one thing, such as the sensor height, and repeat the tests.

The checks last about 30 seconds. Hold one card steady until its reading appears. Run the program again if you need more time. Keep the wheels still.

Teacher check before running

Connect the correct team’s robot. Check the screen Stop control and show how to turn the power off. Raise the wheels for the first check. Keep someone ready to stop the robot.
Keep the wheels still for this lesson. Test the colors before any driving lesson.

LEGO help

Build instructions: https://www.lego.com/service/buildinginstructions/45522

Coding app: https://code.legoeducation.com/

Teacher help: https://teach.legoeducation.com/

Lesson 5: Plan the delivery route

Words to know

- Sequence: Steps in order.
- Calibrate: Test and adjust the settings.
- Variable: One thing you can change, such as drive time.

Coding and testing steps

1. Start with all movement stopped. Plan a short drive, a turn and a second short drive.

2. Move forward slowly for a short time, then stop. Test and adjust that time before adding the next part.

3. Make a short, slow turn using the turning blocks your teacher has checked. The left and right wheels must turn differently to steer. Then stop.

4. Move forward slowly for the second short time, then stop and end the program.

5. For the first tests, keep the total time spent moving to 3 seconds or less. Test each part on the floor. Change one time setting at a time.

Teacher check before running

Connect the correct team's robot. Check the screen Stop control and show how to turn the power off. Raise the wheels for the first check. Keep someone ready to stop the robot. Keep the lane clear and leave extra space beyond the finish.
Start slowly. Use a short time limit and check that the program stops by itself. Stop at once if it moves the wrong way.

LEGO help

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Lesson 6: Stop at the red line

Words to know

- If-then rule: If this happens, do that.
- Loop: Steps that repeat.
- Bounded: Set to stop after a certain time or number of steps.

Coding and testing steps

1. Start with all movement stopped. Set a variable named keepGoing to Yes. A variable stores a value the program can change. Here it remembers whether this test may continue.

2. Repeat the next color-check steps no more than 20 times. Only do them if keepGoing is Yes. If it is No, skip them and do not move.

3. Read the color. If it is white, move forward slowly for 0.1 seconds, then stop.

4. For red or any other reading, stop and set keepGoing to No. Red means delivery. Any other reading means stop and check the setup.

5. After the repeats, stop all movement and end the program. Do not turn keepGoing back to Yes during this test. A person must start a new test.

Teacher check before running

Connect the correct team’s robot. Check the screen Stop control and show how to turn the power off. Raise the wheels for the first check. Keep someone ready to stop the robot. Keep the lane clear and leave extra space beyond the finish.
Start slowly. Use a short time limit and check that the program stops by itself. Stop at once if it moves the wrong way.
Red and unknown readings must end the test. After either reading, the robot must stay stopped even if the sensor later sees white. If the connection is lost, stop using the power-off method you have tested.

LEGO help

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Coding app: https://code.legoeducation.com/

Teacher help: https://teach.legoeducation.com/

Lesson 7: Make the delivery

Words to know

- Success criteria: The things that must happen for a test to count as a success.
- Version: One saved design or program before you change it.
- Reliable: Works as expected again and again.

Coding and testing steps

1. Use the color-stop program from lesson 6. Keep its limit of 20 checks and its short 0.1-second moves.

2. With the power off, attach the checked parcel holder. Put in the same paper parcel used for your earlier tests.

3. Use the white approach lane and red finish line your teacher has checked. Begin close enough to reach the red line within the program’s short time limit.

4. Try three runs from the same start line. Check each rule: parcel stays on; robot stops inside the delivery space; robot hits nothing.

5. If a run does not pass, keep its result in your notes. Change one thing, then try another set of three runs. Do not remove the stop rule to make a delivery work.

Teacher check before running

Connect the correct team’s robot. Check the screen Stop control and show how to turn the power off. Raise the wheels for the first check. Keep someone ready to stop the robot. Keep the lane clear and leave extra space beyond the finish.
Start slowly. Use a short time limit and check that the program stops by itself. Stop at once if it moves the wrong way.
Red and unknown readings must end the test. After either reading, the robot must stay stopped even if the sensor later sees white. If the connection is lost, stop using the power-off method you have tested.

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Module 8: Choose a build and show it

Choose ONE attachment for the same checked robot base. Open worksheets/module-8.pdf for the measured building steps and test tables.

Choice 1: Parcel delivery robot

Use lesson 6's tested color-stop program without removing its limits.

Start stopped; set keepGoing to Yes. Repeat at most 20 times. Only while keepGoing is Yes, read the color: white means a slow 0.1-second move then Stop; any other reading means Stop and set keepGoing to No.

Stop and end after the repeats. Red means delivery; unknown means check the setup.

Choice 2: Drawing robot

Start with movement stopped. Use a checked low speed, such as 20 percent.

Drive forward for 0.8 seconds, then stop and end. Shorten the time if the paper is too small.

No automatic repeat. Lift the marker and return the robot by hand between runs with the power off.

Choice 3: Paper sweeper

Start stopped and set the checked low speed.

Drive forward for 0.8 seconds, then stop and end. Set the collection box using the distance measured in lesson 2.

Do not add continuous driving. Return the paper pieces and robot to their marks between tests.

Choice 4: Ball-pushing robot

Start stopped and use the checked low speed.

Drive forward for 0.8 seconds, then stop and end. Put the goal within the tested travel distance.

No automatic repeat. Return the robot and ball to their marks with the power off.

Choice 5: Rescue carrier

Start stopped and use the checked low speed.

Drive straight for 0.8 seconds, then stop and end. Do not add turns or reversing for this first sled build.

Return the robot and sled to the start with the power off. Do not move a real person, animal or heavy object.

Before driving

Ask the teacher to check the attachment, free wheel movement, sensor view and Stop controls. Turn the power off before changing parts. Try three runs, change one thing and compare. Save the code and explain what happened.